



Operations Research

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Solutions to the Demo Exercise, Sheet 11

Demo Exercises

Exercise D11.1. *Mirror Descent and Bregman Divergence*

Problem

Mirror descent generalizes gradient descent by replacing the Euclidean distance with a Bregman divergence $D_h(x||y)$ generated by a strictly convex function $h(x)$. The update rule in the primal space is given by solving:

$$x^{(k+1)} = \arg \min_x \{ \langle \eta \nabla f(x^{(k)}), x \rangle + D_h(x||x^{(k)}) \}$$

which inherently leads to the dual update: $\nabla h(x^{(k+1)}) = \nabla h(x^{(k)}) - \eta \nabla f(x^{(k)})$.

- Let $h(x) = \frac{1}{2}\|x\|_2^2$. Compute the Bregman divergence $D_h(x||y)$ and show that the Mirror Descent update reduces to standard Gradient Descent.
- Let $h(x) = \sum_{i=1}^n (x_i \log x_i - x_i)$. Compute the gradient $\nabla h(x)$ and its inverse $\nabla h^{-1}(y)$.
- Using the function h from (b), derive the coordinate-wise update rule for $x_i^{(k+1)}$. How does this relate to the Hedge (Exponential Weights) algorithm?

Solution

- For $h(x) = \frac{1}{2}\|x\|_2^2$, the gradient is $\nabla h(x) = x$. The Bregman divergence is defined as:

$$D_h(x||y) = h(x) - h(y) - \langle \nabla h(y), x - y \rangle$$

Substituting h and its gradient:

$$D_h(x||y) = \frac{1}{2}\|x - y\|_2^2$$

Using the dual update $\nabla h(x^{(k+1)}) = \nabla h(x^{(k)}) - \eta \nabla f(x^{(k)})$, we substitute $\nabla h(x) = x$:

$$x^{(k+1)} = x^{(k)} - \eta \nabla f(x^{(k)})$$

This is exactly the standard Gradient Descent update rule.

- (b) For the negative entropy function $h(x) = \sum_{i=1}^n (x_i \log x_i - x_i)$, the partial derivative with respect to x_i is:

$$\frac{\partial h}{\partial x_i} = \log x_i$$

Therefore, the gradient is $\nabla h(x) = (\log x_1, \dots, \log x_n)^T$. The inverse mapping sets $y_i = \log x_i \implies x_i = e^{y_i}$. Thus:

$$\nabla h^{-1}(y) = (e^{y_1}, \dots, e^{y_n})^T$$

- (c) We start with the dual update rule for the i -th coordinate:

$$\nabla h(x^{(k+1)})_i = \nabla h(x^{(k)})_i - \eta \nabla f(x^{(k)})_i$$

Substitute the gradient from (b):

$$\log(x_i^{(k+1)}) = \log(x_i^{(k)}) - \eta \nabla f(x^{(k)})_i$$

Exponentiating both sides yields:

$$x_i^{(k+1)} = x_i^{(k)} e^{-\eta \nabla f(x^{(k)})_i}$$

If we interpret the gradient component $\nabla f(x^{(k)})_i$ as the linear loss $L_i^{(k)}$ incurred by action i , this update rule is exactly the unnormalized multiplicative weight update used in the Hedge (Exponential Weights) algorithm.

Exercise D11.2. Projected Gradient Descent

Problem

Consider minimizing a convex, differentiable function $f(x)$ over $X = \{x \in \mathbb{R}^2 \mid \|x\|_2 \leq 1\}$.

- Formulate the update rule for Projected Gradient Descent using $\Pi_X(\cdot)$.
- Give an explicit formula for the Euclidean projection operator $\Pi_X(y)$ for any point y .
- Let $f(x) = (x_1 - 2)^2 + (x_2 - 2)^2$. Perform one step starting at $x^{(0)} = (0, 0)^T$ with $\eta = 0.5$.

Solution

- $x^{(k+1)} = \Pi_X(x^{(k)} - \eta \nabla f(x^{(k)}))$.
- For the unit disk, the projection scales any vector outside the disk back to the boundary:

$$\Pi_X(y) = \begin{cases} y & \text{if } \|y\|_2 \leq 1 \\ \frac{y}{\|y\|_2} & \text{if } \|y\|_2 > 1 \end{cases}$$

- The gradient is $\nabla f(x) = 2(x_1 - 2, x_2 - 2)^T$. At $x^{(0)} = (0, 0)^T$, $\nabla f(0, 0) = (-4, -4)^T$. The unprojected step is $y^{(1)} = (0, 0)^T - 0.5(-4, -4)^T = (2, 2)^T$. Compute the norm: $\|y^{(1)}\|_2 = \sqrt{8} \approx 2.828 > 1$. Project back to the feasible set X :

$$x^{(1)} = \Pi_X(y^{(1)}) = \frac{1}{\sqrt{8}} \begin{pmatrix} 2 \\ 2 \end{pmatrix} = \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}$$

Exercise D11.3. From Follow-The-Leader to Regularization (FTRL)

Problem

Consider an online linear optimization problem over the decision space $X = [-1, 1]$. In each round t , the adversary reveals a loss function $L_t(x) = g_t x$. The adversary plays the sequence of gradients: $g_1 = 0.5, g_2 = -1, g_3 = 1, g_4 = -1$.

- (a) **The Failure of FTL:** The Follow-The-Leader algorithm chooses $x_t = \arg \min_{x \in X} \sum_{i=1}^{t-1} g_i x$ (with $x_1 = 0$). Compute the sequence of decisions x_t , the cumulative loss of FTL, and the overall regret compared to the best fixed decision in hindsight x^* .
- (b) **The "Cheating" Baseline:** The hypothetical Be-The-Leader (BTL) strategy cheats by looking one step into the future: $x_t^{BTL} = \arg \min_{x \in X} \sum_{i=1}^t g_i x$. Compute the decisions and the regret of the BTL strategy. What does the difference between FTL and BTL reveal about the nature of adversarial regret?
- (c) **The Cure (FTRL):** To prevent the wild oscillations of FTL, we introduce Follow-The-Regularized-Leader with a strongly convex penalty $h(x) = \frac{1}{2}x^2$ and a regularization weight $\eta^{-1} > 0$. The update rule becomes:

$$x_{t+1} = \arg \min_{x \in X} \left(\sum_{s=1}^t g_s x + \frac{1}{2\eta} x^2 \right)$$

Derive the closed-form expression for x_{t+1} as a function of the historical gradients and η . Briefly discuss how this relates to Online Gradient Descent (OGD) and prevents the failure observed in part (a).

Solution

(a) FTL Evaluation:

- $t = 1$: $x_1 = 0$. Observe $g_1 = 0.5$. Loss: 0.
- $t = 2$: $\sum g = 0.5 \implies x_2 = -1$. Observe $g_2 = -1$. Loss: 1.
- $t = 3$: $\sum g = -0.5 \implies x_3 = 1$. Observe $g_3 = 1$. Loss: 1.
- $t = 4$: $\sum g = 0.5 \implies x_4 = -1$. Observe $g_4 = -1$. Loss: 1.

FTL Cumulative Loss = 3.

The sum of all gradients is -0.5 . The best fixed decision minimizes $-0.5x$, so $x^* = 1$. The cumulative loss of x^* is -0.5 .

Regret = $\text{Loss}_{FTL} - \text{Loss}_{x^*} = 3 - (-0.5) = 3.5$.

(b) BTL Evaluation (Cheating):

- $t = 1$: $\text{sum} = 0.5 \implies x_1^{BTL} = -1$. Loss: -0.5 .
- $t = 2$: $\text{sum} = -0.5 \implies x_2^{BTL} = 1$. Loss: -1 .
- $t = 3$: $\text{sum} = 0.5 \implies x_3^{BTL} = -1$. Loss: -1 .

- $t = 4$: $\text{sum} = -0.5 \implies x_4^{BTL} = 1$. Loss: -1 .

BTL Cumulative Loss $= -3.5$.

BTL Regret $= -3.5 - (-0.5) = -3.0$.

Insight: BTL achieves strictly negative regret. This reveals that FTL's failure is solely due to the "delay" of reacting one step too late. The adversary exploits this by making FTL oscillate exactly out of phase with the optimal choices.

(c) **FTRL Derivation:** Setting the derivative of the regularized objective to zero:

$$\sum_{s=1}^t g_s + \frac{1}{\eta} x = 0 \implies x = -\eta \sum_{s=1}^t g_s$$

Projecting onto the bounded space $X = [-1, 1]$:

$$x_{t+1} = \Pi_{[-1,1]} \left(-\eta \sum_{s=1}^t g_s \right)$$

Insight: Since $\sum_{s=1}^t g_s = \sum_{s=1}^{t-1} g_s + g_t$, this update is mathematically equivalent to an Online Gradient Descent step followed by projection. By managing the step length via η , FTRL limits rapid structural movements across the boundaries, neutralizing adversarial exploitation to achieve sublinear $O(\sqrt{T})$ regret bounds.